

Ian Zhang

Curriculum Vitae

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EDUCATION

PhD in Statistics

University of Toronto

Sep 2026-Present

- Advisors: Joshua Speagle and Thibault Randrianarisoa

MSc in Statistics

University of Toronto

Sep 2025-Jun 2026

- Advisor: Nancy Reid

Honours BSc in Statistics & Mathematics

University of Toronto

Sep 2021-Jun 2025

- Advisor: Jun Young Park
- Graduated with high distinction

PUBLICATIONS AND PREPRINTS

†: Corresponding author; *: Equal contribution

Published

Markov Processes for Enhanced Deepfake Generation and Detection

2026

Michael A. Kouritzin, Ian Zhang[†], Jyoti Bhadana, Seoyeon Park

AIMS Mathematics, 11(4): 11731–11759. doi: [10.3934/math.2026483](https://doi.org/10.3934/math.2026483)

Under Review

Parametric Hotspot Analysis with the Getis-Ord Statistic

2026

Aditya Khan, Ian Zhang

In Progress

Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks

2026

Ian Zhang[†], Thibault Randrianarisoa

Redundancy-Aware Model Assessment for Component and Cluster Selection in Clinical Subtype Discovery

2026

Ian Zhang^{,†}, Hanlong Chen^{*}, ..., Howard Chertkow, ..., Malcolm Binns*

Continuous Cognitive Heterogeneity in Dementia: A Severity Factor and an Age-Related Amnesic Axis Without Reproducible Subtypes

2026

Hanlong Chen^{}, Ian Zhang^{*}, ..., Malcolm Binns*

Spatial Dependence Regularization for Improved Generalization in Climate Downscaling Models

2026

Ian Zhang[†], Finn Tran, Meredith Franklin

**LiBRE: Statistical Learning for Representative Brine Design and Lithium
Enrichment Analysis**

2026

*Gabriel Liu**, *Ian Zhang**[†], ..., *Arbav Nambiar*, ..., *Daniel Alessi*

EXPERIENCE

Graduate Research Assistant

2025-Present

Baycrest Institute/University of Toronto

- Supervisor: Malcolm Binns

Summer Research Trainee

2025

McGill University

- Supervisor: [Jianguo \(Jeff\) Xia](#)

Research Assistant

2023-2024

University of Alberta

- Supervisor: [Mike Kouritzin](#)

TEACHING

Teaching Assistant

2025–2026 School Year

STA302: Methods of Data Analysis I

Fall 2025

MAT137: Calculus with Proofs

Fall 2025 & Winter 2026

STA437: Methods of Multivariate Data

Winter 2026

UofT Statistical Sciences Research Program (UTSSRP)

Summer 2026

2026–2027 School Year

MAT148: Calculus I with Proofs

Fall 2026

STA257: Probability and Statistics I

Fall 2026

MAT149: Calculus II with Proofs

Winter 2027

STA457: Time Series Analysis

Winter 2027

PRESENTATIONS

Conference Talks

2026 SSC Annual Meeting in Hamilton

2026

Presented *Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks* in the New Frontiers in Bayesian Analysis session.

2025 SSC Annual Meeting in Saskatoon

2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Probability session.

Third Joint SIAM/CAIMS Annual Meetings

2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Machine Learning and Stochastic Processes session.

Posters

The Thirteenth Annual Canadian Statistics Student Conference

2025

Presented *Comparing Exact and Approximate Inference for Spatial Autocorrelation Tests: A Power Analysis*.

AWARDS

Anna & Alex Beverly Memorial Fellowship

2025

Awarded by University College (University of Toronto) for general proficiency during my undergraduate degree.

NSERC Undergraduate Student Research Award (x2)

2023, 2024

Awarded by the University of Alberta to research Markov process methods for deepfake generation and detection.

Jack Zwaigenbaum Scholarship

2022

Awarded by University College (University of Toronto) for academic excellence in my first year of undergraduate studies.

Dean's List Scholar (x4)

2022, 2023, 2024, 2025

Awarded by the Faculty of Arts and Science (University of Toronto) for maintaining academic excellence every year of my undergraduate degree.

GRANTS

CSSC Travel Grant

2025

Awarded to full-time Canadian students presenting a poster or giving a talk at the Canadian Statistics Student Conference.

SERVICES

Departmental Services

VP-Academic, Department of Statistical Sciences (University of Toronto)

2026-2027

Conferences

Session Chair, Third Joint SIAM/CAIMS Annual Meetings

2025

Chaired the "Machine Learning and Stochastic Processes" session.

OTHER

Citizenship: Canadian

Languages Spoken: English (native), Mandarin (native), Japanese (intermediate)

Memberships: Statistical Society of Canada